

Foundation (Jalapeno)

- Q1.** To convert $x^2 + 6x + 8$ to vertex form, what value is added and subtracted inside the parentheses?
- (A) 8
(B) 9
(C) 3
(D) 4
(E) 2
- Q2.** What is the vertex form of $x^2 + 4x + 4$?
- (A) $(x + 2)^2$
(B) $(x - 2)^2$
(C) $(x + 4)^2$
(D) $(x - 4)^2$
(E) $(x + 1)^2$
- Q3.** The score s in a golf game can be modeled by $s = x^2 + 2x - 8$. Convert this to vertex form by completing the square.
- (A) $s = (x + 1)^2 - 9$
(B) $s = (x - 1)^2 - 9$
(C) $s = (x + 2)^2 - 12$
(D) $s = (x - 2)^2 - 12$
(E) $s = (x + 4)^2 - 16$
- Q4.** If the distance d thrown by a shot putter is given by $d = t^2 - 4t + 4$, what is the vertex form of this equation?
- (A) $d = (t - 2)^2$
(B) $d = (t + 2)^2$
(C) $d = (t - 2)^2 - 4$
(D) $d = (t + 2)^2 + 4$
(E) $d = (t - 4)^2 + 16$
- Q5.** The number of basketball free throws f made after p practices can be modeled by $f = p^2 + 2p + 1$. What is the vertex form of this equation?
- (A) $f = (p - 1)^2$
(B) $f = (p + 1)^2$
(C) $f = (p - 2)^2$
(D) $f = (p + 2)^2$
(E) $f = (p + 3)^2$
- Q6.** What is the x -coordinate of the vertex for the equation $x^2 + 8x + 7$ after completing the square?
- (A) -4
(B) -3
(C) -2
(D) -1
(E) 0
- Q7.** A volleyball player's height h in centimeters after t years is given by $h = t^2 + 6t + 10$. Convert this equation to vertex form.
- (A) $h = (t + 3)^2 + 1$
(B) $h = (t - 3)^2 + 1$
(C) $h = (t + 3)^2 - 1$
(D) $h = (t - 3)^2 - 1$
(E) $h = (t + 2)^2 + 6$
- Q8.** Which equation represents the vertex form of $x^2 - 2x + 3$ after completing the square?
- (A) $x^2 - 2x + 1 + 2$
(B) $x^2 - 2x + 2 + 1$
(C) $(x - 1)^2 + 2$
(D) $(x + 1)^2 + 2$
(E) $(x - 2)^2 + 3$
- Q9.** The number of basketball free throws f made after p practices can be modeled by $f = p^2 + 2p + 1$. What is the vertex form of this equation?
- (A) $f = (p - 1)^2$
(B) $f = (p + 1)^2$
(C) $f = (p - 2)^2$
(D) $f = (p + 2)^2$
(E) $f = (p + 3)^2$

Open Ended Questions (FRQ)

- Q9.** Convert the equation $y = x^2 + 10x + 24$ to vertex form and identify the vertex. Show all steps.
- Q10.** The cost c of producing n items can be modeled by $c = n^2 + 4n + 5$. Complete the square to convert this equation into vertex form, and then interpret the results in the context of cost and production.

Chef's Tips

- Provide a review of completing the square before the test.
- Ensure students have graph paper for questions requiring graphical interpretation.
- Emphasize the practical applications of vertex form in sports and fitness.